Week 8: STRUCTURAL CONTEXTS

What are the contexts in which subjectivities link with broader social structures and relations, and manifest in maps, map-making, and spatial data visualization?

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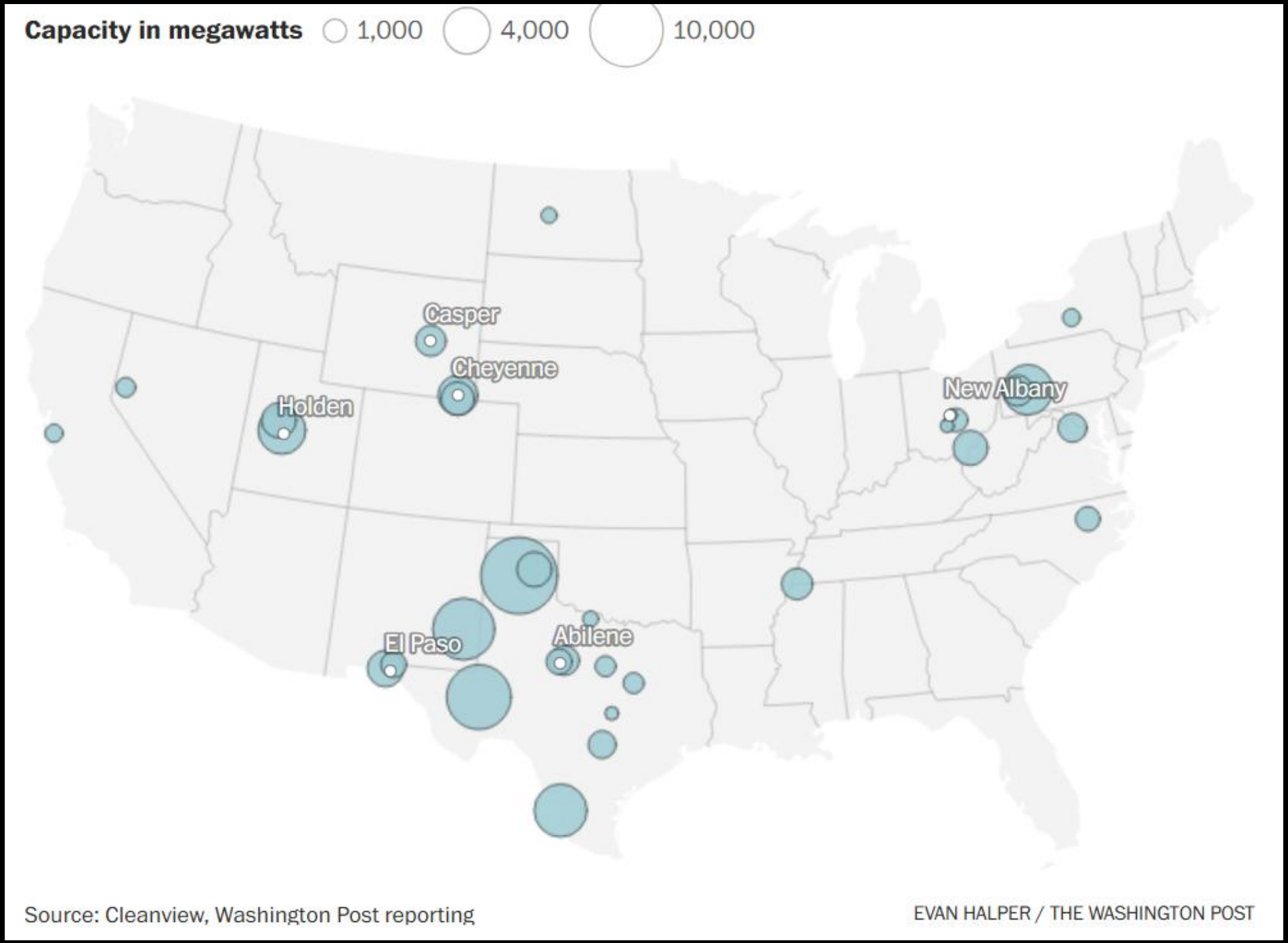
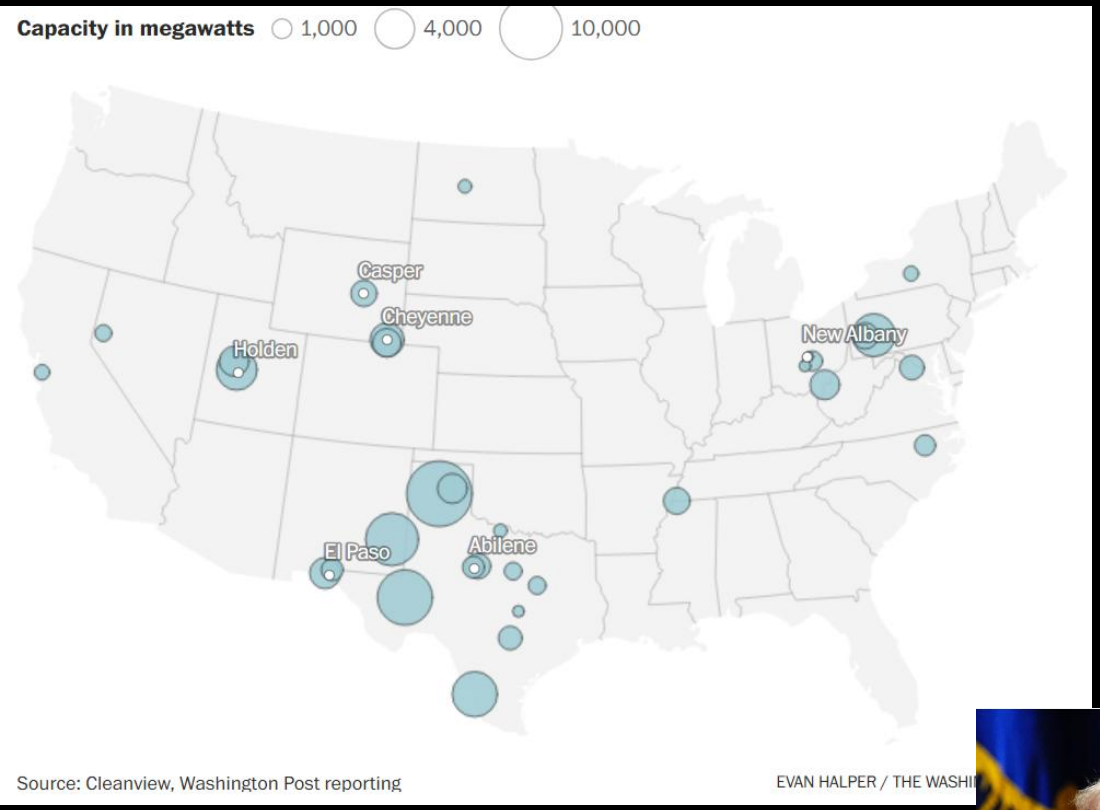


https://www.washingtonpost.com/business/2026/02/19/data-centers-power-grid-ai/



https://www.washingtonpost.com/business/2026/02/19/data-centers-power-grid-ai/

Week 8: STRUCURAL CONTEXTS

Read:

- **M Chs 6-10.**
- **Anderson, Jung, "Extending the conversation on socially engaged geographic visualization".**

Mapping practice #4: Social Contexts

“[M]aps, by articulating the world in mass-produced and stereotyped images, express an embedded social vision. Consider, for example, the fact that the ordinary road atlas is among the best selling paperback books in the United States and then try to gauge how this may have affected ordinary Americans' perception of their country. What sort of an image of America do these atlases promote?
JB Harley, “Deconstructing the Map”

“A map says to you, ‘Read me carefully, follow me closely, doubt me not.’ It says, ‘I am the earth in the palm of your hand. Without me, you are alone and lost.’”
Beryl Markham, *West with the Night*

structural/social context:

what is social context?
(re)interpretations of data
case: food deserts

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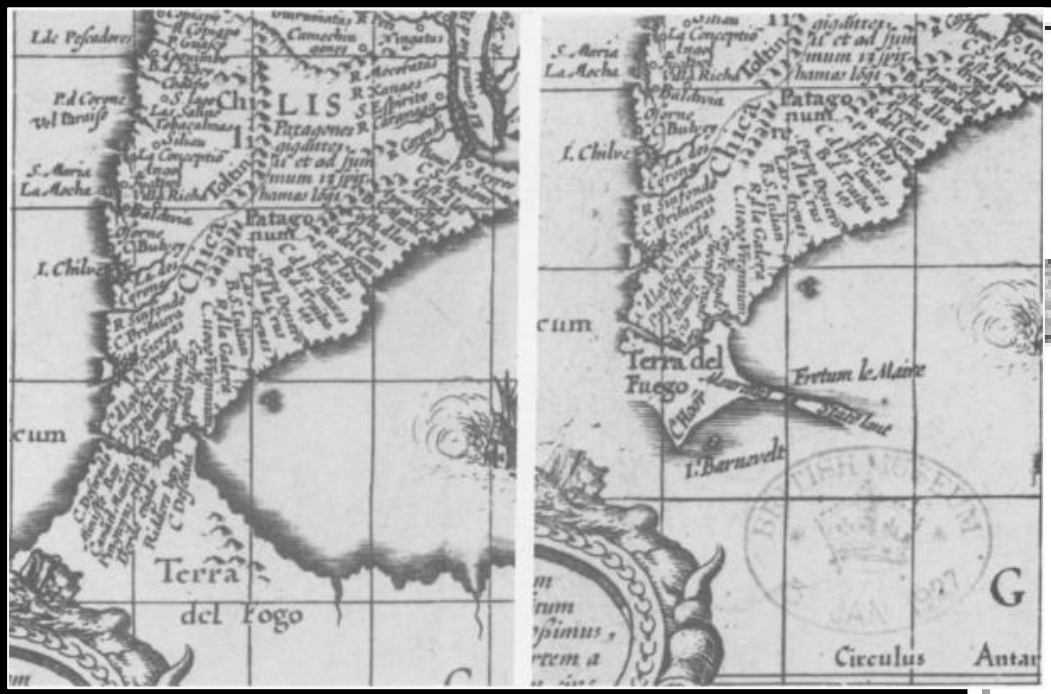
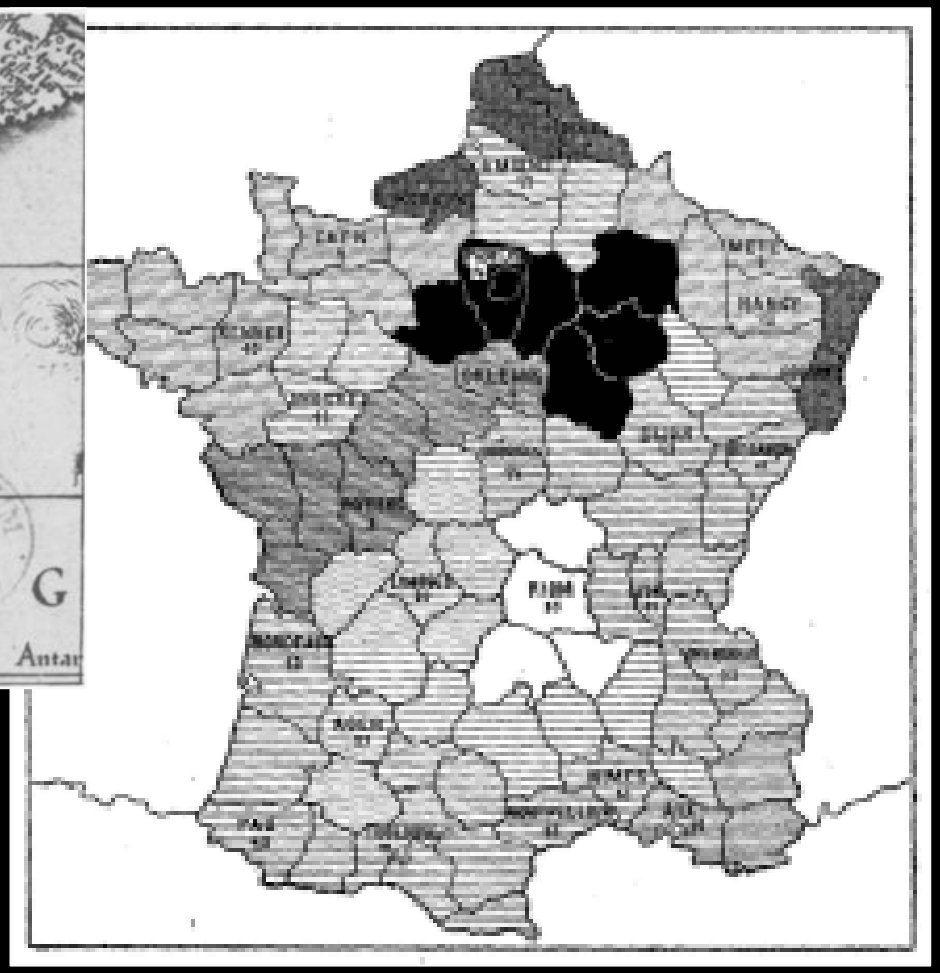
what do we mean by social context?

https://www.facebook.com/photo?fbid=1529783088510394&set=a.610868297068549



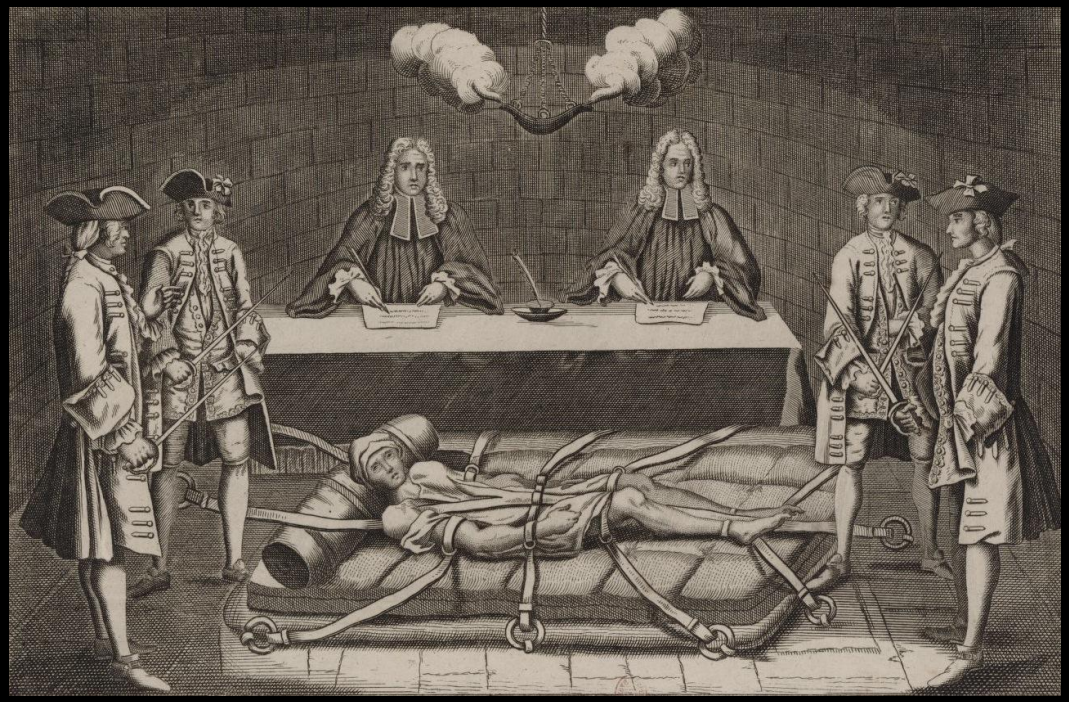
state secrets protecting commercial interests

maps and the state



first known choropleth map (by charles dupin, 1826)

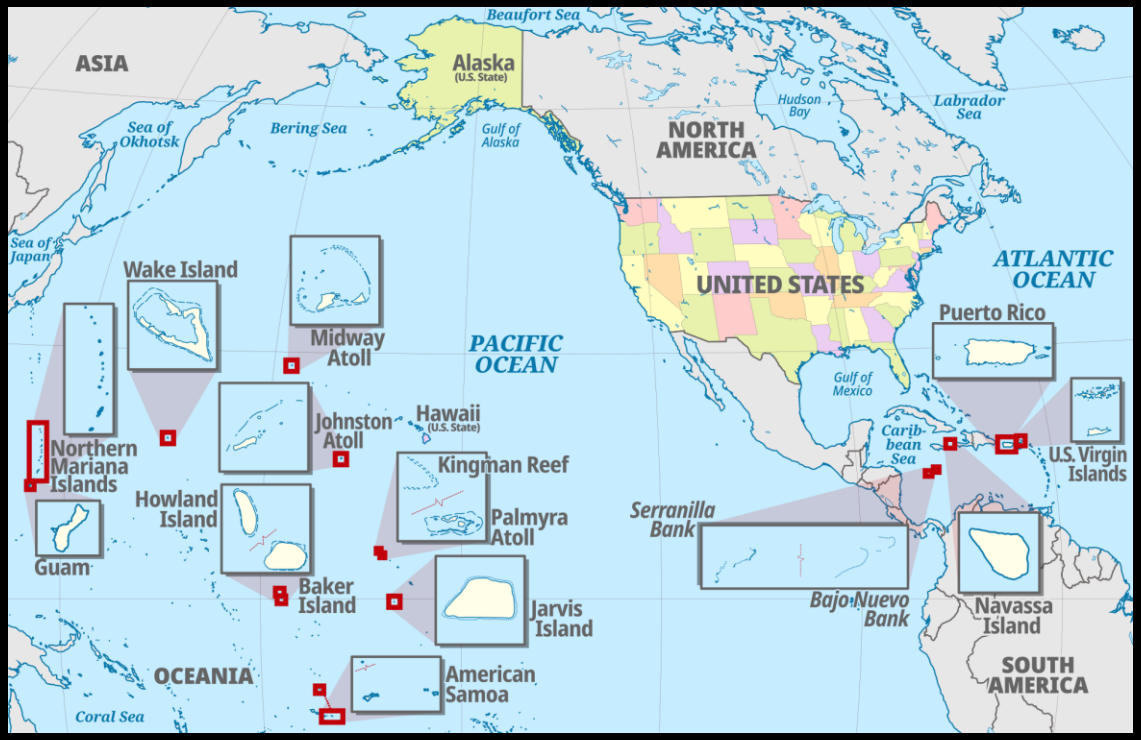
power over individuals → management of populations



‘sovereign’ map

queen elizabeth I (first country atlas)

https://en.wikipedia.org/wiki/Territories_of_the_United_States#/media/File:United_States_(+overseas),_administrative_divisions_-_en_-_colored_(zoom).svg



‘colonial’ map

institutional contexts

"Gambling, for instance, is linked to higher rates of suicide. In the Canadian province of Alberta in 2003,there were 140 suicides linked to gambling debts. In Ontario in 2003, there were 15 deaths classified as suicides that were linked to gambling. Ontario's population is four times that of Alberta, and presumably there are proportionally more gambling venues. Should one conclude that gambling is more profitable in Ontario than Alberta and thus leads to fewer cases of depression linked to debt? A more likely scenario is that the criteria for linking suicides togambling are more stringent in Ontario than Alberta."
Nadine Schuurman, "Database Ethnographies…"

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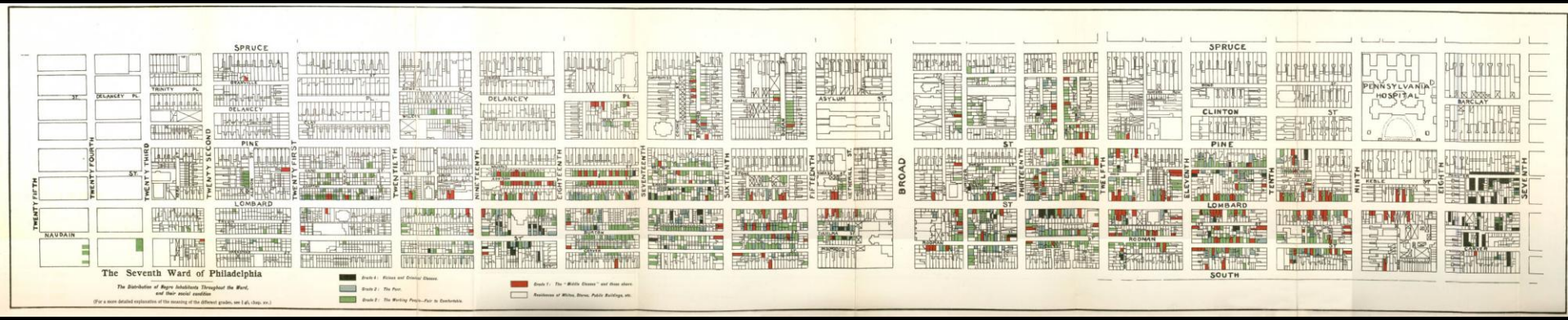
(re)interpretations of data

case: food deserts

data: selections, interpretations



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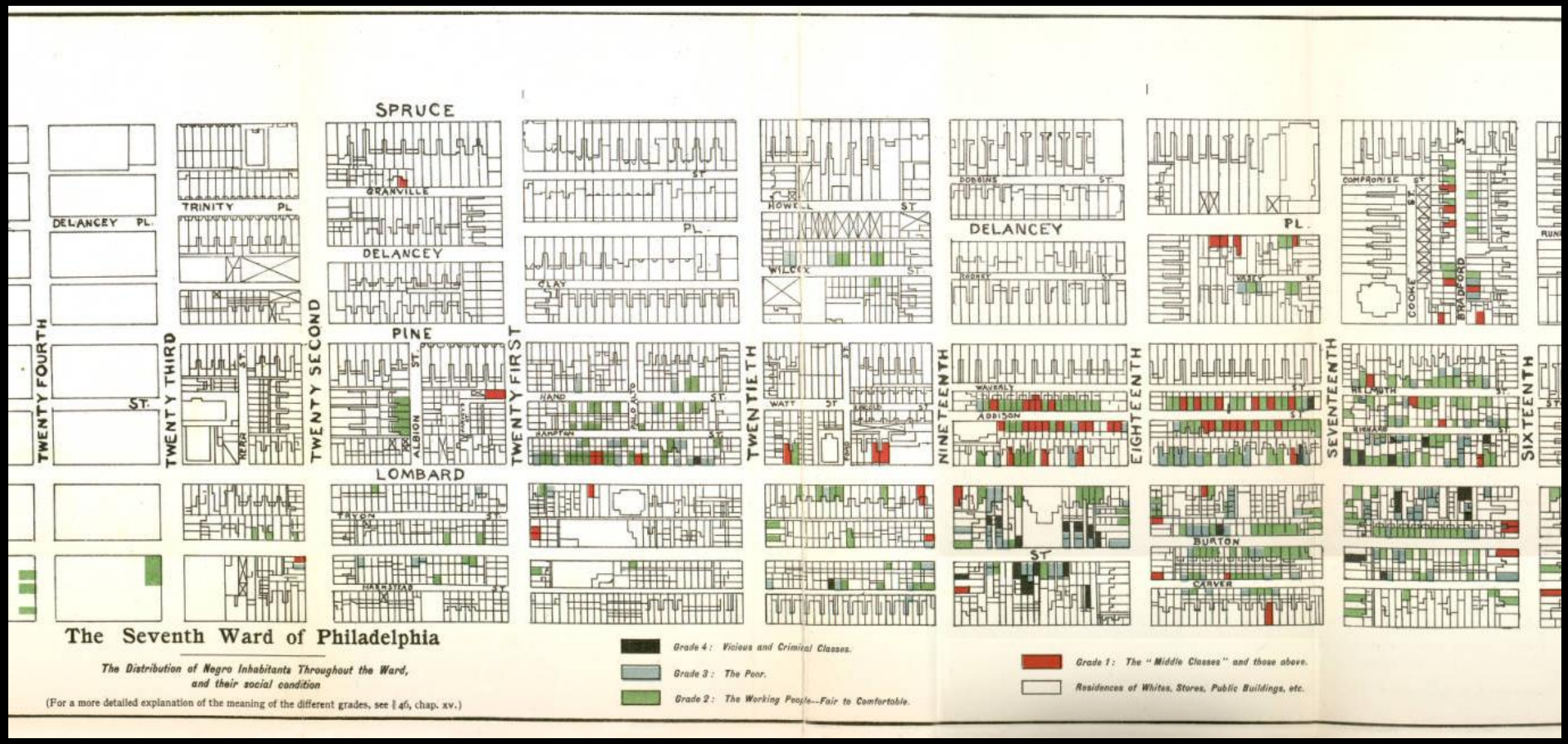
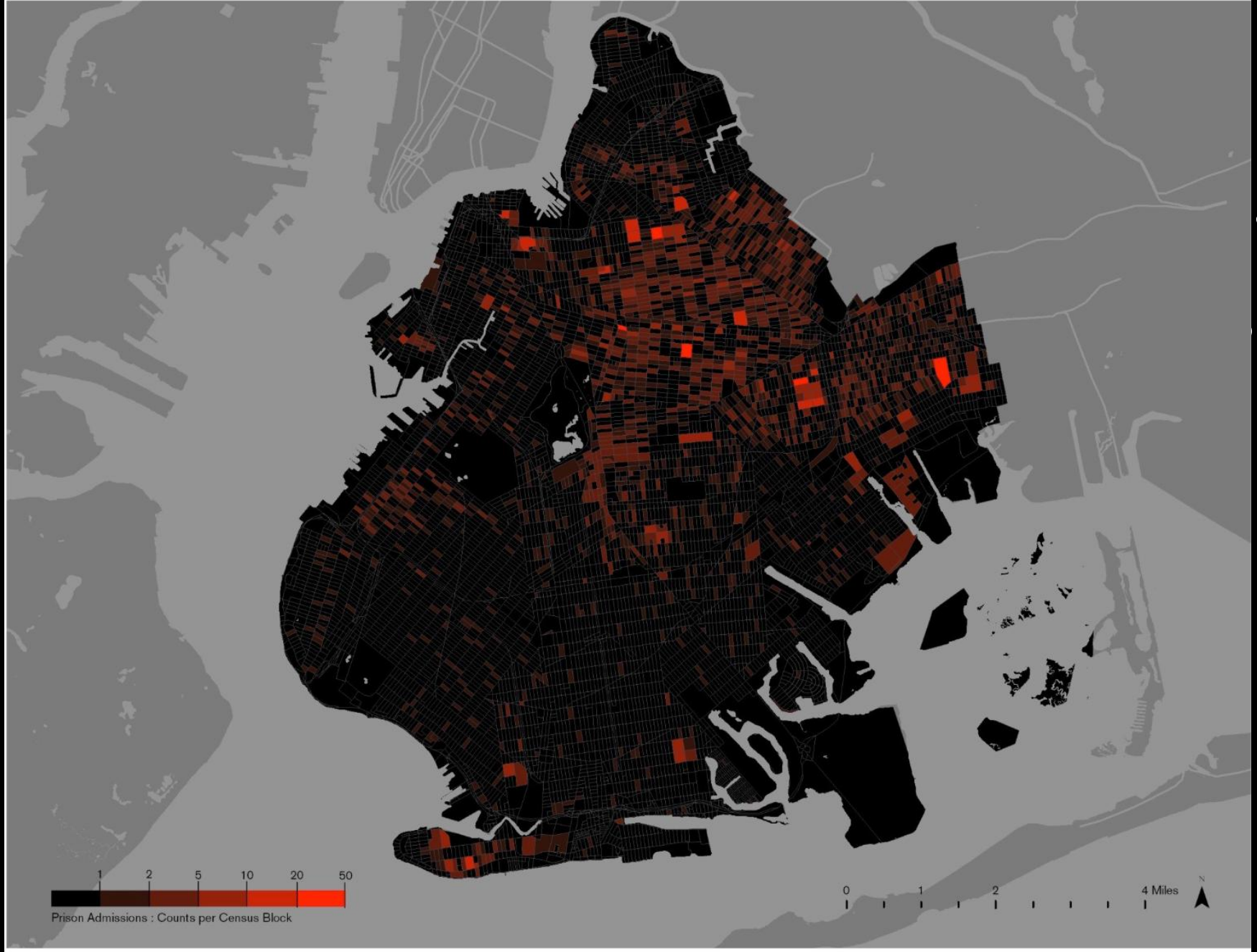
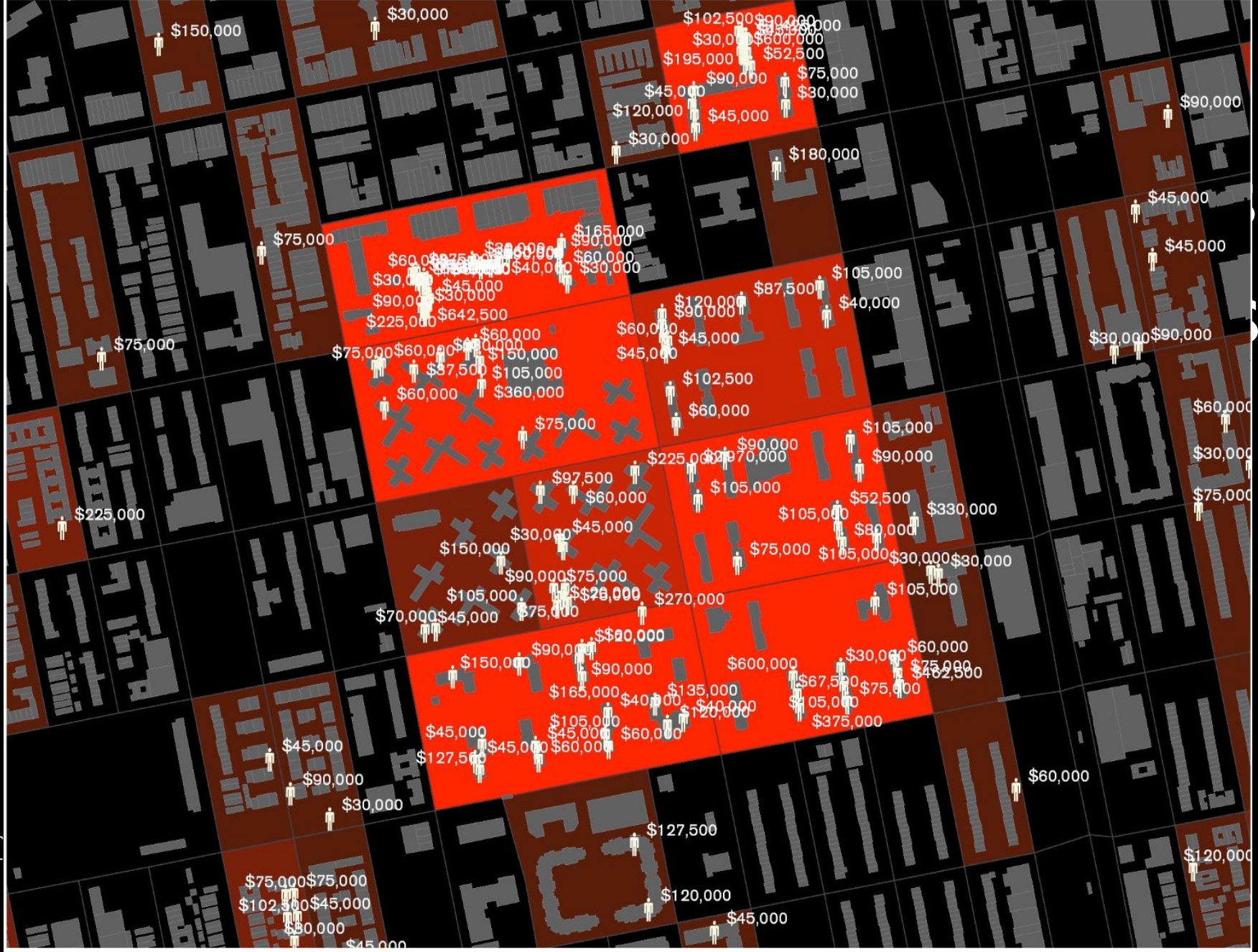


https://exhibits.temple.edu/s/william-still/page/w-e-b--du-bois-seventh-ward-ma



Data in geographic context shows people in prision are highly concentrated in specific neighborhoods.

https://c4sr.columbia.edu/projects/million-dollar-blocks



It cost $11,839,665 to incarcerate people from these 11 blocks in 2003. We call these Million Dollar Blocks.

https://c4sr.columbia.edu/projects/million-dollar-blocks

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Drug Discovery Today
Volume 30, Issue 4, April 2025, 104346

Post-screen (grey)

Drug development for neglected ultra-rare diseases of no commercial interest: Challenges and opportunities

Joseph A. Katakowski [1], Juan C. López [2]

Show more

Around the time Microsoft invested $10 billion in OpenAI, it laid off ten thousand workers to cut costs. After Google watched OpenAI outpace it, it centralized its AI labs into Google DeepMind. As Baidu raced to develop its ChatGPT equivalent, employees working to advance AI technologies for drug discovery had to suspend their research and cede their computer chips to develop the chatbot instead. The current AI paradigm is also choking off alternative paths to AI development. The number of independent researchers not affiliated with or receiving funding from the tech industry has rapidly dwindled, diminishing the diversity of ideas in the field not tied to short-term commercial benefit. Companies themselves, which once invested in sprawling exploratory research, can no longer afford to do so under the weight of the generative AI development bill. Younger generations of scientists are falling in line with the new status quo to make themselves more employable. What was once unprecedented has become the norm. -Karen Hao, *Empires of AI*

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case: food deserts

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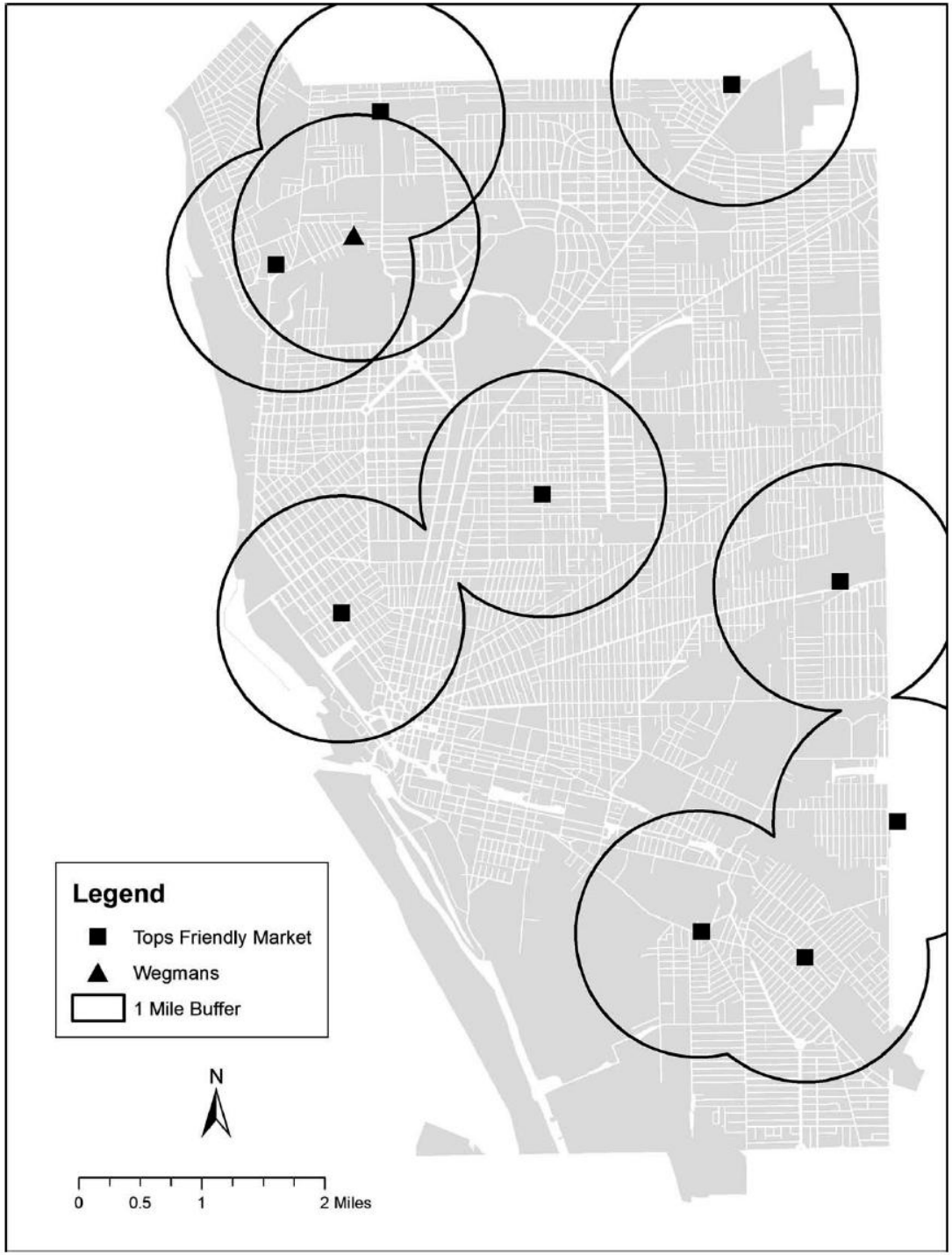
Census tracts qualify as food deserts if they meet low-income and low-access thresholds:

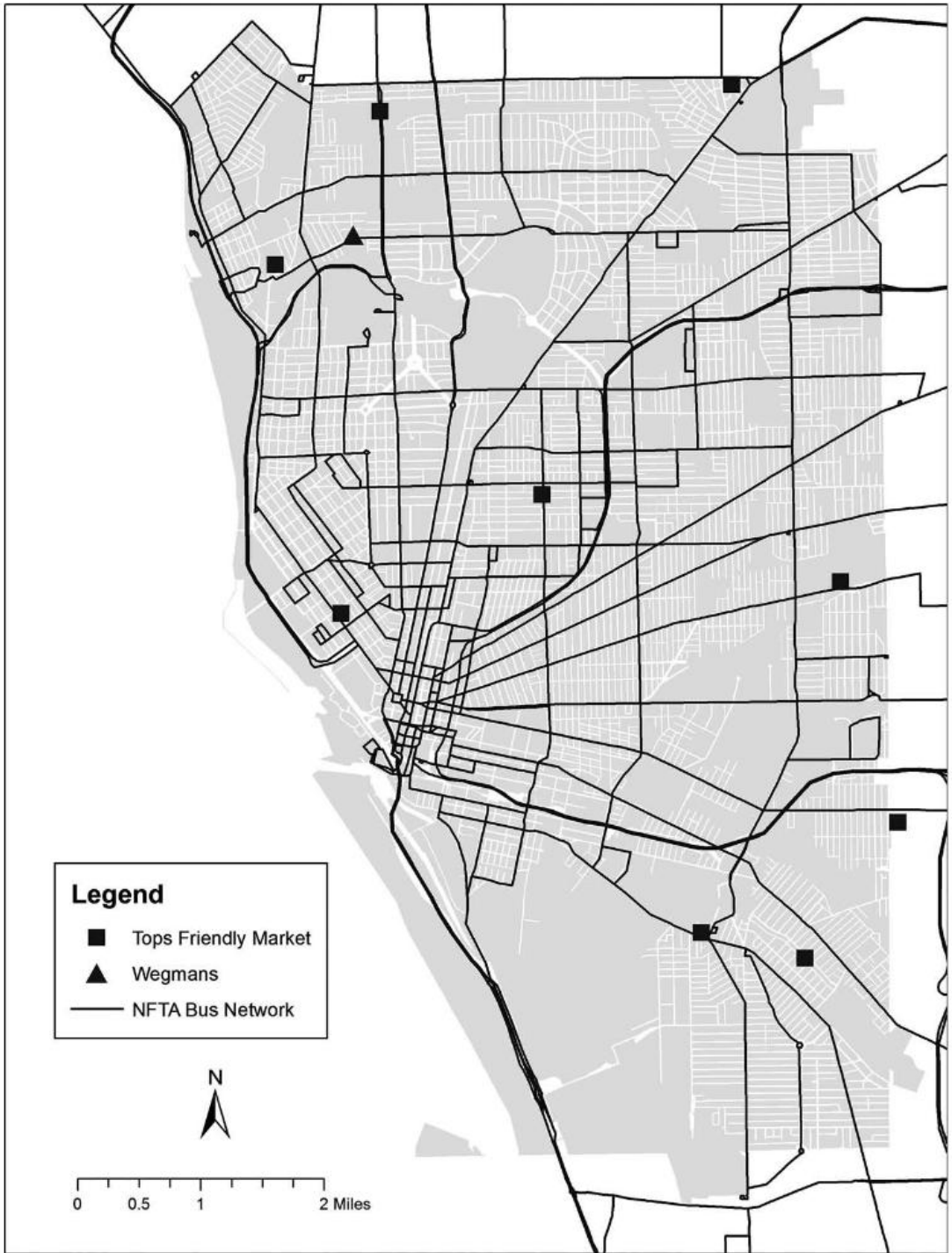
•Low-income: a **poverty rate of 20 percent or greater**, or a median family income at or below 80 percent of the statewide or metropolitan area median family income;

•Low-access: **at least 500 persons and/or at least 33 percent of the population lives more than 1 mile** from a supermarket or large grocery store (10 miles, in the case of rural census tracts).

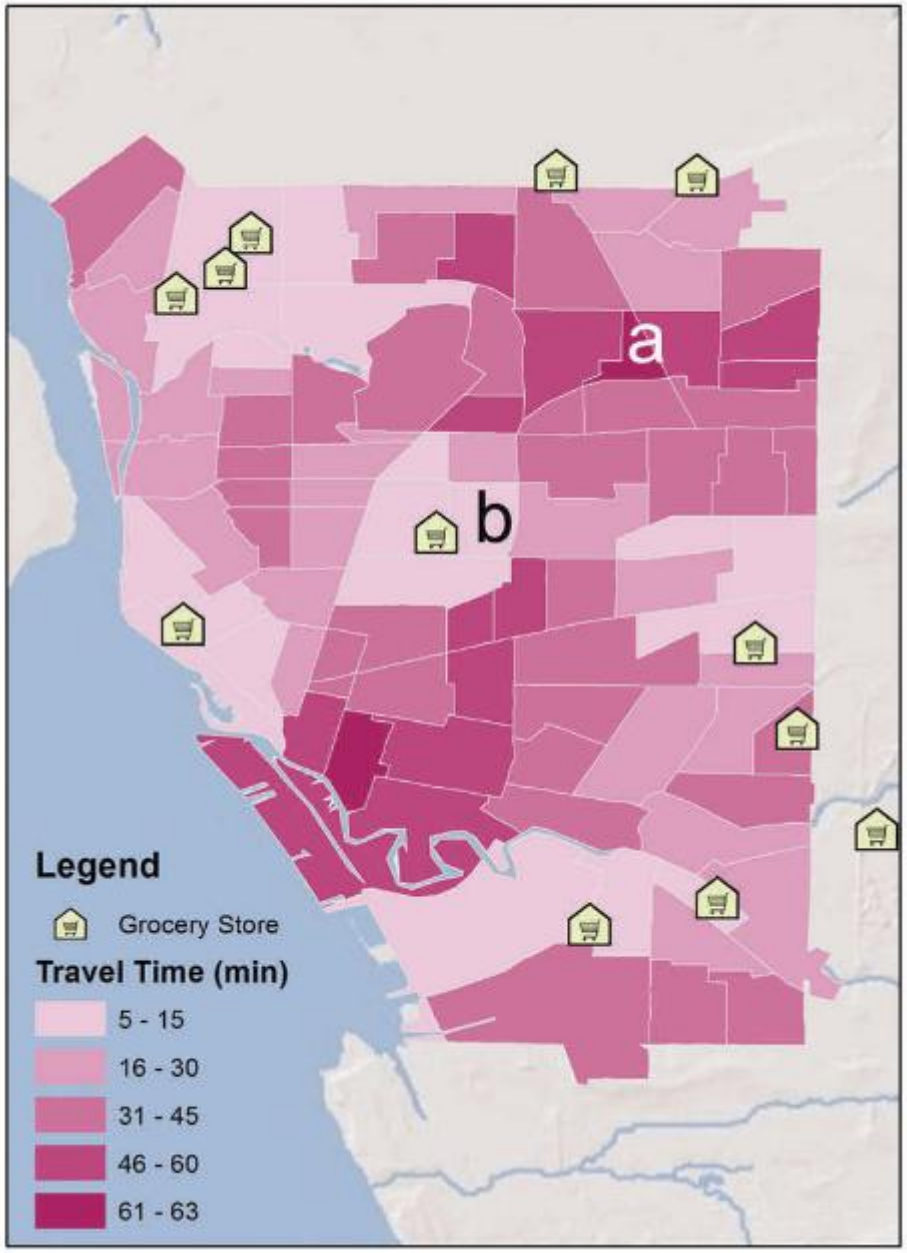
-USDA (https://www.ers.usda.gov/amber-waves/2011/december/data-feature-mapping-food-deserts-in-the-u-s)

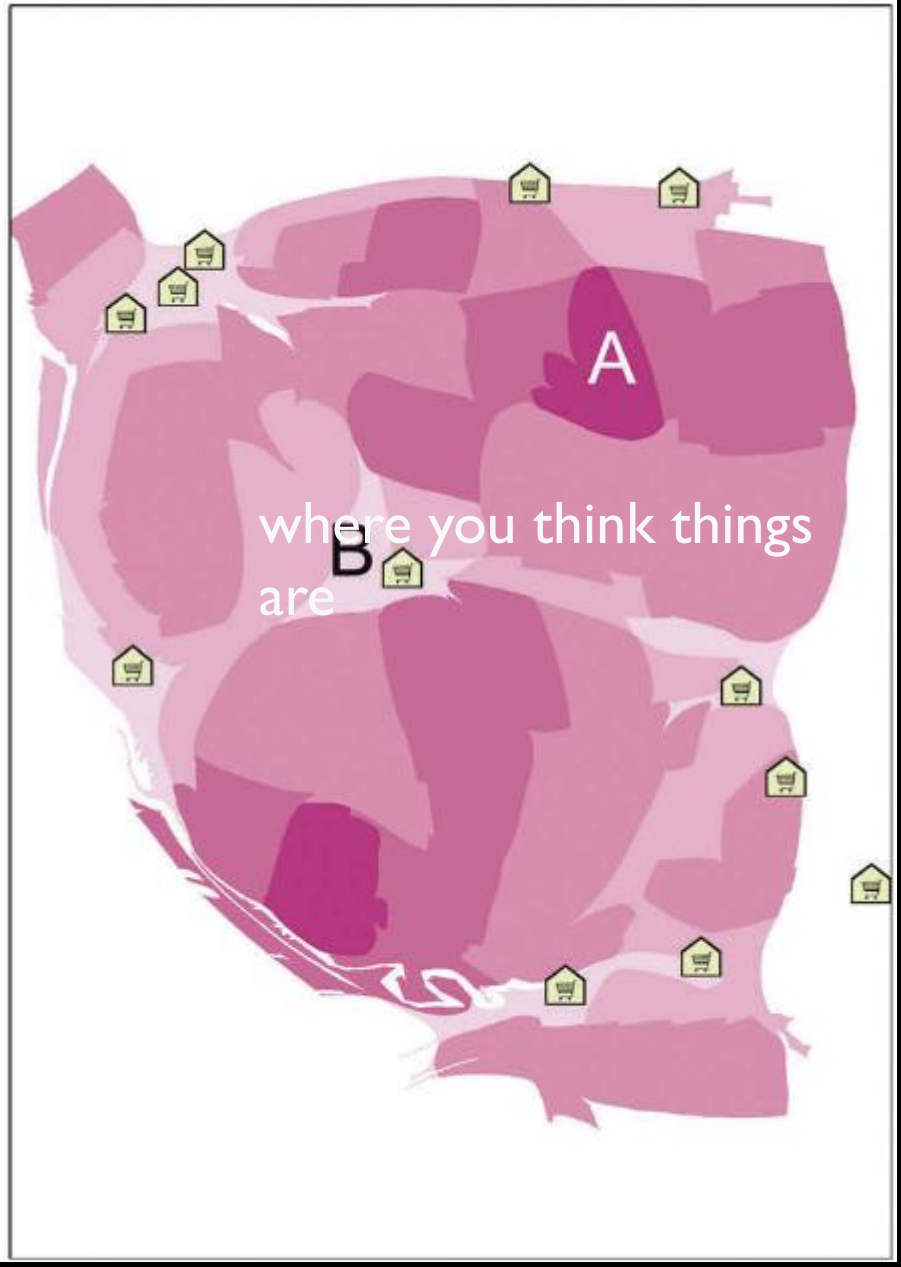


case: food deserts



food deserts: transit


a
b
Legend
Grocery Store
Travel Time (min)
5 - 15
16 - 30
31 - 45
46 - 60
61 - 63


A
B
where you think things are